Tucker ALS for Functional Approximation in Chebyshev Bases

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1 Problem Setup

Given a smooth function $f : \mathbb{R}^3 \to \mathbb{R}$ sampled on a tensor grid, we approximate it using a Tucker representation built directly from Chebyshev basis matrices.
Let

$$T_x \in \mathbb{R}^{N_x \times (d_x+1)}, \quad T_y \in \mathbb{R}^{N_y \times (d_y+1)}, \quad T_z \in \mathbb{R}^{N_z \times (d_z+1)}$$

be Chebyshev Vandermonde matrices evaluated at the grid points.
We seek the approximation

$$F \approx (T_x A_x,\ T_y A_y,\ T_z A_z,\ G),$$

where $A_x \in \mathbb{R}^{(d_x+1)\times R_x}$, $A_y \in \mathbb{R}^{(d_y+1)\times R_y}$, $A_z \in \mathbb{R}^{(d_z+1)\times R_z}$ are factor matrices, and $G \in \mathbb{R}^{R_x \times R_y \times R_z}$ is the core tensor.

2 ALS Algorithm

The ALS loop alternates updates over the three modes, each optimizing one factor matrix and the core tensor while fixing the others.
Define

$$U_x = T_x A_x, \quad U_y = T_y A_y, \quad U_z = T_z A_z.$$

2.1 Mode–1 Update

We rewrite F with the x-mode singled out via mode-1 unfolding:

$$F_{(1)} \approx U_x\, G_{(1)} (U_z \otimes U_y)^T,$$

where $F_{(1)} \in \mathbb{R}^{N_x \times (N_y N_z)}$.
Let $YZ = U_z \otimes U_y \in \mathbb{R}^{(N_y N_z)\times(R_y R_z)}$. We solve

$$\min_{\tilde{G}_x} \|F_{(1)} - T_x \tilde{G}_x\, YZ^T\|_F.$$

Vectorizing and solving the least-squares problem:

$$\mathrm{vec}(\tilde{G}_x) = (YZ \otimes T_x)^\dagger\, \mathrm{vec}(F_{(1)}),$$

yielding $\tilde{G}_x \in \mathbb{R}^{(d_x+1)\times(R_y R_z)}$.

Apply SVD and truncate:

$$\tilde{G}_x = U_x^c \, S_x \, V_x^T, \quad A_x = U_x^c(:, 1:R_x), \quad G_{(1)} = S_x(1:R_x, 1:R_x) \, V_x(1:R_x, :).$$

Reshape $G_{(1)}$ back into the core tensor G.

2.2 Mode–2 and Mode–3 Updates

The mode-2 and mode-3 updates proceed analogously, working with the respective mode unfoldings $F_{(2)}$ and $F_{(3)}$, and updating A_y, A_z, and the core G accordingly.
Iterate through all three modes until convergence.

3 Extension to Higher Dimensions

For a d-dimensional function with Chebyshev Vandermonde matrices

$$T_k \in \mathbb{R}^{N_k \times (m_k+1)}, \qquad k = 1, \ldots, d,$$

we seek

$$F \approx (T_1 A_1, \; T_2 A_2, \; \ldots, \; T_d A_d, \; G),$$

with $A_k \in \mathbb{R}^{(m_k+1) \times R_k}$ and $G \in \mathbb{R}^{R_1 \times \cdots \times R_d}$.
The mode-n update solves

$$\mathrm{vec}(F_{(n)}) = (U_{\neg n} \otimes T_n) \; \mathrm{vec}(\tilde{G}_n),$$

where $U_{\neg n} = U_d \otimes \cdots \otimes U_{n+1} \otimes U_{n-1} \otimes \cdots \otimes U_1$ and $U_k = T_k A_k$.
SVD and truncation yield $A_n = U_n^c(:, 1:R_n)$ and an updated core G. Cycle through $n = 1, \ldots, d$ until convergence.

4 Numerical Experiments

We test the algorithm on 2D anisotropic functions of the form

$$f(x, y) = \frac{1}{1 + \|C[x, y]^T\|^2}, \quad (x, y) \in [-1, 1]^2,$$

where $C \in \mathbb{R}^{2 \times 2}$ controls the anisotropy and orientation.
Case 1: $C = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$ (horizontal stretching).
Case 2: $C = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} R(30°)$ (rotated vertical stretching).

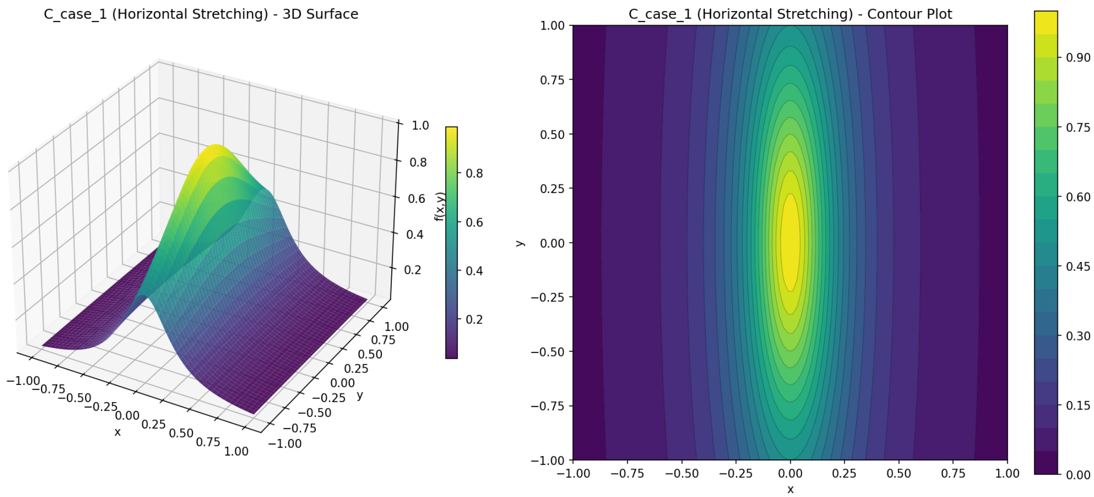

Figure 1: Case 1 test function.

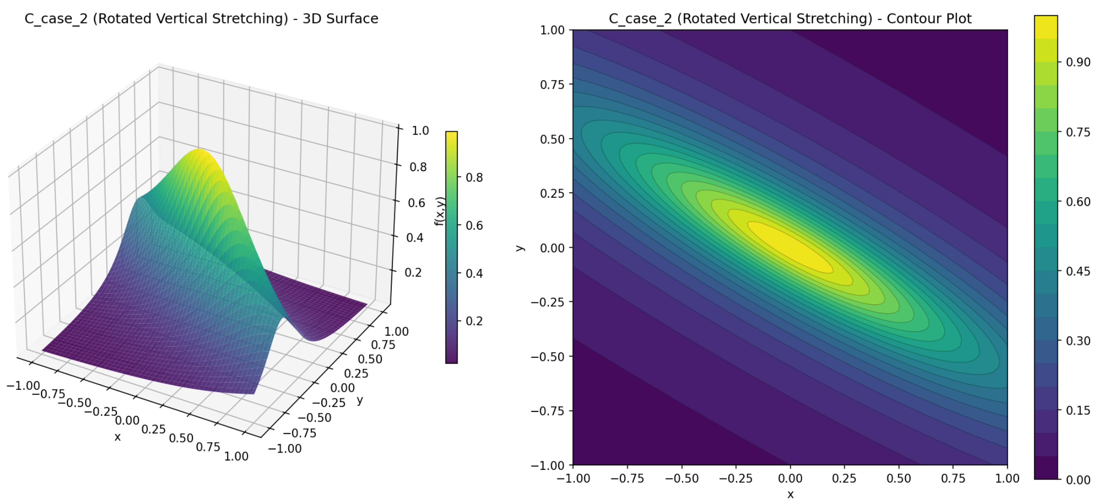

Figure 2: Case 2 test function.

4.1 Approximation Quality

We measure the ℓ^2 coefficient error between the Chebyshev coefficient tensor reconstructed from each method's Tucker approximation at a given rank and the best degree-d Chebyshev interpolant. Comparisons include:

ALS Tucker: The direct ALS algorithm described above.

Tucker Baseline: Two-stage approach: compute full Chebyshev interpolant, then apply Tucker decomposition.

ADAM/BFGS: Gradient-based optimization methods.

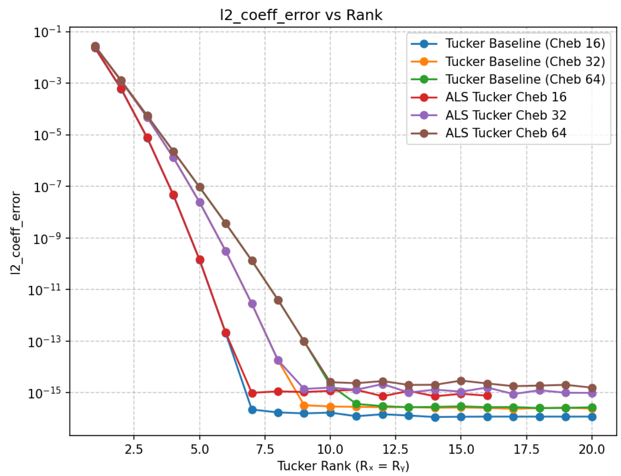

Figure 3: Case 1: Coefficient error vs. Tucker rank for degrees $d = 16, 32, 64$ on Chebyshev grids.

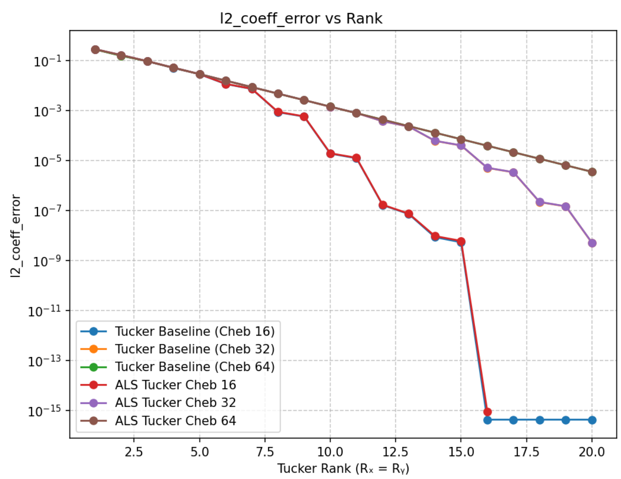

Figure 4: Case 2: Coefficient error vs. Tucker rank for degrees $d = 16, 32, 64$ on Chebyshev grids.

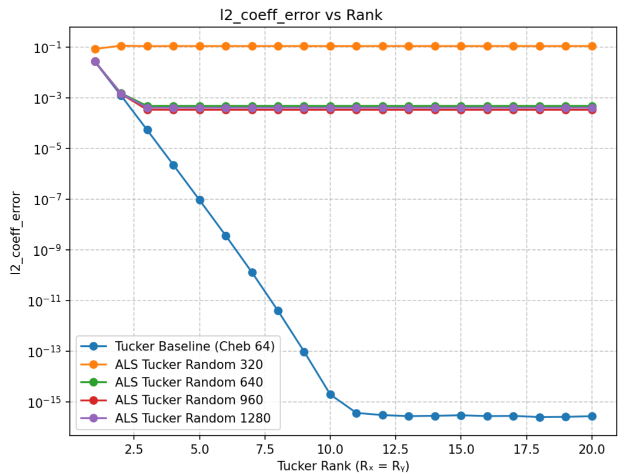

Figure 5: Case 1: Coefficient error using 320×320 random sampling points.

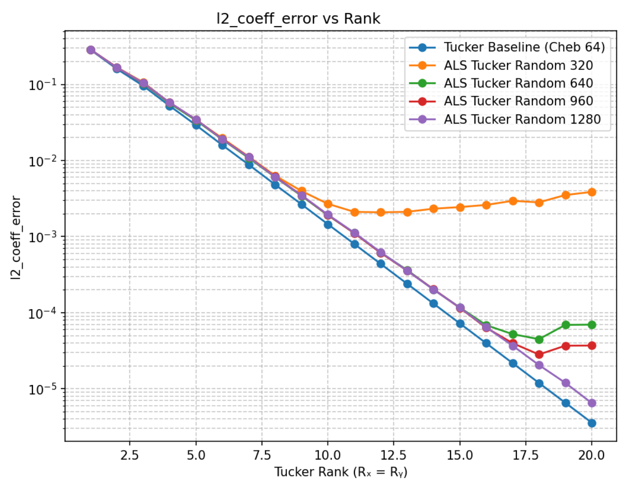

Figure 6: Case 2: Coefficient error using 320×320 random sampling points.

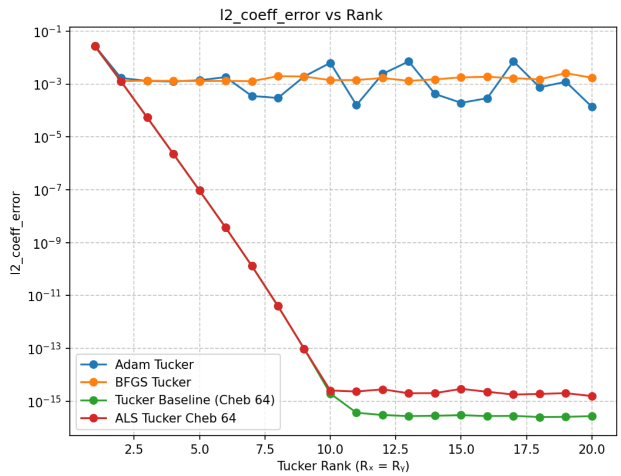

Figure 7: Case 1: Comparison of ALS Tucker, Tucker Baseline, ADAM, and BFGS optimization methods.

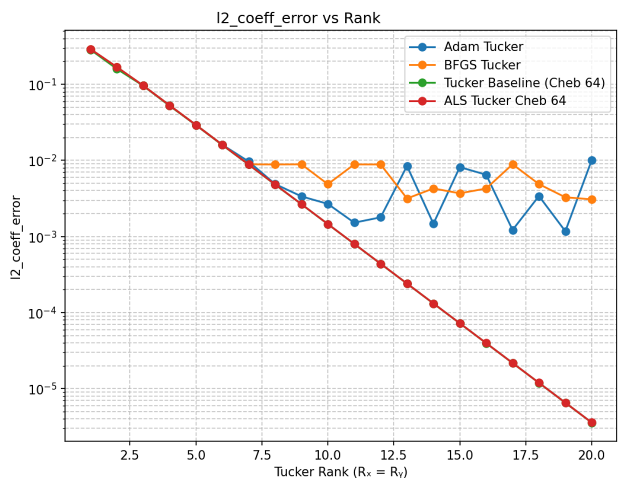

Figure 8: Case 2: Comparison of ALS Tucker, Tucker Baseline, ADAM, and BFGS optimization methods.

5 Comparison of Tucker Decomposition Methods

For a 2D function sampled on an $N_x \times N_y$ grid, we compare three Tucker-based approaches focusing on the mode-1 update:

Aspect	**Classical Tucker**	**SVD-based ALS (Ours)**	**Direct ALS**
Factorization	$F \approx U_x G U_y^T$ where $U_x \in \mathbb{R}^{N_x \times R_x}$, $U_y \in \mathbb{R}^{N_y \times R_y}$	$F \approx (T_x A_x) G (T_y A_y)^T$ where $A_x \in \mathbb{R}^{(d_x+1)\times R_x}$, $A_y \in \mathbb{R}^{(d_y+1)\times R_y}$	$F \approx (T_x A_x) G (T_y A_y)^T$ where $A_x \in \mathbb{R}^{(d_x+1)\times R_x}$, $A_y \in \mathbb{R}^{(d_y+1)\times R_y}$
Variables	Factor matrices U_x, U_y directly	Coefficient matrices A_x, A_y	Coefficient matrices A_x, A_y
Mode-1 update	$F \approx U_x G U_y^T$ $M = F U_y G^T$ $U_x = \mathrm{SVD}_{R_x}(M)$	$F \approx T_x A_x G (T_y A_y)^T$ $\min \lVert F - T_x \tilde{G}_x (T_y A_y)^T \rVert_F$ $\tilde{G}_x = [(T_y A_y) \otimes T_x]^\dagger \mathrm{vec}(F)$ $\tilde{G}_x = U^c S V^T$ (SVD) $A_x = U^c(:, 1\!:\!R_x), G = S(1\!:\!R_x, :)V^T$	$F \approx T_x A_x G (T_y A_y)^T$ $\min \lVert F - T_x A_x G (T_y A_y)^T \rVert_F$ $G_x = T_x^T T_x,\ H_y = A_y^T T_y^T T_y A_y$ $G_x A_x (G H_y G^T) = T_x^T F T_y A_y G^T$ $A_x = G_x^{-1} [T_x^T F T_y A_y G^T][G H_y G^T]^\dagger$
d-dim extension (mode-n)	$U_n \leftarrow \mathrm{SVD}_{R_n}(F_{(n)} U_{\neg n} G_{(n)}^T)$ where $U_{\neg n} = \bigotimes_{k \neq n} U_k$	$\tilde{G}_n = [U_{\neg n} \otimes T_n]^\dagger \mathrm{vec}(F_{(n)})$ $\tilde{G}_n = U^c S V^T$ (SVD) $A_n = U^c(:, 1 : R_n), G_{(n)} = S(1\!:\!R_n, :)V^T$	$G_n A_n M_n = T_n^T F_{(n)} (\bigotimes_{k \neq n} T_k A_k) G_{(n)}^T$ where $M_n = G_{(n)} (\bigotimes_{k \neq n} H_k) G_{(n)}^T$ $H_k = A_k^T (T_k^T T_k) A_k$ for $k \neq n$
Rank adaptation	Fixed rank R_x, R_y	Rank adapted via SVD truncation at each iteration	Fixed rank R_x, R_y
Parameters	$N_x R_x + N_y R_y + R_x R_y$	$(d_x + 1) R_x + (d_y + 1) R_y + R_x R_y$	$(d_x{+}1) R_x {+} (d_y{+}1) R_y {+} R_x R_y$
Interpretability	No functional meaning	Chebyshev coefficients	Chebyshev coefficients

Table 1: Comparison of three Tucker decomposition approaches for 2D functions.